Yeshan Qiu

Tel: +65 8038 5367 | Email: E0788063@u.nus.edu | yeshan.qiu@sec.ethz.ch

EDUCATION

National University of Singapore (NUS), Singapore 01/2022–2026

PhD in Architecture

PhD awarded October 2026.

Dissertation: Understanding Impact of Urban Patterns on Urban Avian Biodiversity: Comparative Studies Across Cities

Shanghai Jiao Tong University (SJTU), Shanghai, China 09/2018–03/2021

Master's Degree in Landscape Architecture

Dissertation: Construction and Design Application of Bayesian Model of Public Ecological Aesthetic Perception in Urban Green Space Based on Deep Learning

Hainan University (HNU), Haikou, China 09/2014–06/2018

Bachelor's Degree in Landscape Architecture

Overall GPA: 3.82/4.0 | Ranking: 1/136

RESEARCH INTERESTS

- Urban biodiversity, avian ecology, and biodiversity modelling
- Climate change impacts, climate × land-use interactions, and cross-city comparative ecology
- Urban landscape patterns, 3D vegetation structure, and pattern–process relationships

PUBLICATIONS

Manuscripts under review / in development

***Qiu, Y.**, Chi, D., Hahs, A. K., Richards, D. R., Yang, J., Meili, N., Zhu, Y., Wang, J., Lin, B. B., Manoli, G., Zhang, Y., Burlando, P., McDonnell, M. J., & Tan, P. Y. (2026). Systematic Review of the Relationships between Multidimensional Urban Patterns and Avian Biodiversity: A Meta-analysis. [Under Review]*

***Qiu, Y.**, Chi, D., Hahs, A. K., Richards, D. R., Yang, J., Meili, N., Zhu, Y., Wang, J., Lin, B. B., Manoli, G., Zhang, Y., Burlando, P., McDonnell, M. J., & Tan, P. Y. (2026). Mapping Methods Matter: Influence of Spatial Analytical Units on Urban Avian Biodiversity Assessment: A Meta-analysis. [Under Review]*

***Qiu, Y.**, Chi, D., Hahs, A. K., Richards, D. R., Yang, J., Meili, N., Zhu, Y., Wang, J., Lin, B. B., Manoli, G., Zhang, Y., Burlando, P., McDonnell, M. J., & Tan, P. Y. (2026). Assessing Urban Pattern Effects on Bird Species Richness Across 68 Cities. [Under Review]*

Peer-reviewed journal articles

***Qiu, Y.**, Pan, H., Kalantari, Z., Giusti, M., & Che, S. (2023). The natural focus: Combining deep learning and eye-tracking to understand public perceptions of urban ecosystem aesthetics. Ecological Indicators, 156, 111181.*

***Qiu, Y.**, Che, S. (2022). Construction Strategies for Urban Ecological Green Space Based on Ecological Aesthetics. Landscape Architecture, 29(7), 37–43.*

***Qiu, Y.**, Xie, C., Pan, H., & Che, S. (2021). Study of Urban Street-Scene and Landscape Aesthetic Public Perception Based on Deep Learning: The Case of Shanghai. Chinese Landscape Architecture, 37(6), 77–81.*

*Chi, D., Manoli, G., Lin, B., Aerts, R., Yang, J., Hahs, A., Richards, D., Meili, N., **Qiu, Y.**, et al. (2025). Residential tree canopy configuration and mortality in 6 million Swiss adults: a longitudinal study. The Lancet Planetary Health, 9(3), e186–e195.*

*Meili, N., Zheng, X., Takane, Y., Nakajima, K., Yamaguchi, K., Chi, D., Zhu, Y., **Qiu, Y.**, et al. (2025). Modeling the effect of trees on energy demand for indoor cooling and dehumidification across cities and climates. Journal of Advances in Modeling Earth Systems, 17(3), e2024MS004590.*

*Zhu, Y., Burlando, P., Zhang, Y., Huang, J., Chi, D., Wang, J., **Qiu, Y.**, Manoli, G., Fatichi, S., & Tan, P. Y. (2026). Associations between urban form and urban multifunctionality: a comparative study. npj Urban Sustainability. https://doi.org/10.1038/s42949-026-00481-3*

*Zhu, Y., Wang, J., Zhang, Y., Hwang, Y. H., Chi, D., **Qiu, Y.**, Chen, X., Feng, C., Zhou, Z., Huang, J., Burlando, P., & Tan, P. Y. (2026). Research-informed tools as a science-practice interface for mediating multifunctional design: reflections from a collaborative design research studio. Socio-Ecological Practice Research. https://doi.org/10.1007/s42532-026-00273-0*

*Chi, D., Richards, D., Manoli, G., Yang, J., Ji, J. S., Zhang, Y., Lin, B., Hahs, A., Liu, C., Zhu, Y., **Qiu, Y.**, Wang, J., Burlando, P., Fatichi, S., & Tan, P. Y. (2026). Urban tree canopy fragmentation and child mortality in low- and middle-income countries. Proceedings of the National Academy of Sciences, 123(34), e2606669123. https://doi.org/10.1073/pnas.2606669123*

*Chi, D., Manoli, G., Yang, J., Richards, D., Hahs, A., Lin, B., McDonnell, M. J., Zhang, Y., Zhu, Y., **Qiu, Y.**, Wang, J., Zheng, X., Burlando, P., Fatichi, S., & Tan, P. Y. (2026). Tree canopy configuration and Swiss adult mortality at the municipal level: A nationwide ecological study. Environment International, 209, 110188. https://doi.org/10.1016/j.envint.2026.110188*

*Zhu, Y., Wang, J., Manoli, G., Zhang, Y., Chi, D., Meili, N., **Qiu, Y.**, Lin, G. S., Tan, P. Y., Burlando, P., & Fatichi, S. (2025). Influence of urban form and function on daytime-nighttime population differences and hazard risk assessments. npj Urban Sustainability, 5, 91.*

*Sun, W., **Qiu, Y.**, Yu, B., & Wang, Y. (2024). Inclusive Design of Smart Landscape in Community Parks for The Migrant Elderly. Human Factors in Design, Engineering, and Computing, 159(159).*

***Qiu, Y.**, Xu, X. (2020). Landscape Design for Modern Healing Spaces under the Influence of Traditional Chinese Health Culture. Chinese Horticulture Abstracts, 33(7), 106–108.*

Preprints / Working Papers

*Wang, J., Zhang, Y., Huang, W., Chi, D., Zhu, Y., **Qiu, Y.**, Burlando, P., & Tan, P. Y. (2026). How urban form shapes small business density: Global evidence from 80 cities. SSRN preprint 6748941. https://doi.org/10.2139/ssrn.6748941*

CONFERENCE PRESENTATIONS AND ABSTRACTS

Qiu, Y. et al. (2026). Sampling bias in citizen-science bird data. TDWG 2026 Annual Conference, Oslo, Norway. Oral presentation. Best Student Oral Presentation Award.

Qiu, Y., Chi, D., Zhu, Y., Wang, J., McDonnell, M. J., & Tan, P. Y. (2025). Thinking Vertically: How 3D Urban Form Shapes Global Avian Biodiversity: Multi-City evidence for functional guild-specific responses to 3D urban form. British Ecological Society Annual Meeting 2025. Poster presentation.

Qiu, Y., Chi, D., Meili, N., Zhu, Y., Wang, J., McDonnell, M. J., & Tan, P. Y. (2024). Exploring the Impact of Urban Pattern on Avian Occurrence: A Buffer-Based Analysis Using Citizen Science Data. AGU Fall Meeting 2024. Poster presentation / conference abstract.

Qiu, Y., Chi, D., Meili, N., Zhu, Y., Wang, J., McDonnell, M. J., & Tan, P. Y. (2024). Exploring the Impact of Urban Pattern Metrics on Avian Occurrence: A Buffer-Based Analysis Using Citizen Science Data. Ecological Society of America Annual Meeting 2024. Poster presentation.

Meili, N., Zheng, X., Chi, D., Zhu, Y., **Qiu, Y.**, Wang, J., Manoli, G., Burlando, P., et al. (2023). Quantifying Vegetation Effects on Climate and Building Energy Demand in Cities. AGU Fall Meeting Abstracts, GC44F-01.

Chi, D., Meili, N., Zhu, Y., Wang, J., **Qiu, Y.**, Li, C., Burlando, P., Fatichi, S., Lin, B., et al. (2024). Associations between greenspace patterns and mortality in six million Swiss adults: a longitudinal study. ISEE Conference Abstracts, 1.

Qiu, Y., Chen, Y., & Che, S. (2020). Using Eye-tracking and Deep Learning Approach to Promote Naturalness in Urban Landscape. EGU General Assembly Conference Abstracts, EGU2020-6515.

Chen, Y., **Qiu, Y.**, & Che, S. (2020). Combining deep learning and multi-source data to promote subjective perception of ecosystem services in urban landscape. EGU General Assembly Conference Abstracts, EGU2020-6516.

RESEARCH EXPERIENCE

Comparative Ecology of Cities: Understanding Patterns, Processes and Properties of Cities to Improve Urban Conditions 03/2023–Present

Future Cities Lab Global, NUS, ETH Zurich | *PhD Researcher*

• Contributed to a collaborative, multi-module project examining urban pattern–process–function relationships across cities, linking settlement patterns with biodiversity, climate and energy, hydrology, public health, and socioeconomic outcomes.

• Independently led the biodiversity module through a cross-city comparative study of 68 global cities, integrating citizen-science species occurrence records, biodiversity observation data, and spatial pattern metrics into modelling-ready datasets to assess relationships between urban spatial patterns and biodiversity responses.

• Conducted multi-source, multi-city urban pattern metrics analysis, integrating land-use/land-cover data, urban spatial layers, landscape metrics, and 3D urban/vegetation structure indicators.

• Analysed how landscape distribution, composition, configuration, and three-dimensional structure shape biodiversity responses, with implications for planning practices, ecological networks, and nature-based solutions.

Collaborative Urban Bird Modelling Projects with NParks 07/2024–Present

Wildlife Management Research, National Parks Board, Singapore | *Researcher*

• Collaborated with NParks' Wildlife Management Research team on applied avian biodiversity modelling projects, including the Decadal Study of Birds in Singapore and straw-headed bulbul population modelling.

• Integrated bird survey data with Landsat-derived land-cover layers, LiDAR-derived vegetation structure metrics, ETH green canopy data, and multi-scale landscape metrics.

• Generated modelling-ready spatial predictors from transect buffers, survey points, and prediction grids, including landscape composition, configuration, canopy connectivity, patch structure, total edge, core area, and percentage cover metrics.

• Supported long-term urban bird community analysis and straw-headed bulbul abundance/population modelling for management-oriented biodiversity assessment and urban wildlife planning.

Public Risk Perception, Behaviour Patterns and Public Emotional Guidance under Major Public Health Emergencies 03/2020–2023

National Natural Science Youth Fund Project | *Researcher*

• Analysed the impact and potential role of urban green space on public health events and public health.

• Concluded that group pressure and psychological restoration of elderly residents in the community were strongly correlated with green-space rate and green-view measures.

Synergistic Research on Street Species and High-Quality Road Greening in Shanghai 11/2019–09/2020

Shanghai Science and Technology Commission | *Team Leader*

• Responsible for the evaluation of benefits of urban street greening in Shanghai using AI deep learning skills.

• Assisted in optimization of ecological design, including increasing the plant landscape (after evaluation) by 23.2%, seasonal colour richness of plant landscape by 33.67%, and overall green space utilization by 8%.

• Led 6 team members to implement preliminary investigations and assessments of ecological road environments.

• Designed a natural plant community pattern in urban road green space across the Jing'an district of Shanghai and independently produced a research report.

• Utilised ecological natural plant-community modes to replace traditional geometric and regular plant communities in road green spaces to improve ecological stability and increase biodiversity.

Survey of Shanghai Urban Landscape and Style: Artificial Intelligence and Big Data 03/2019–11/2020

National Natural Science Youth Fund Project | *Research Assistant*

• Interviewed 86 households face-to-face to investigate public perception of urban landscape aesthetics, using deep learning, street data, and eye-tracking technology.

• Analysed street-view data with Python to explore 238 road landscapes in downtown Shanghai.

• Set up a street-view sample database to conduct eye-movement experiments on 57 local residents in Shanghai and tracked saccades, gazes, glances, and other physiological indicators individually.

• Built an aesthetic cognition Bayesian model using deep learning to expand limited aesthetic survey samples, aiming to construct a rapid and large-scale survey of public aesthetic preferences.

• Published research outputs in *Chinese Landscape Architecture*.

PROFESSIONAL EXPERIENCE

Dawson Landscape Design Co., Ltd. 01/2018–08/2018

Landscape Designer Assistant

• Participated in preliminary analysis of the status quo in beautiful village revitalization construction planning and design projects in Haikou city.

• Independently designed Ding'an Community Park planning projects within two weeks and produced the first and final floor plans, which were adopted by clients.

• Collected data on site history, topography, precipitation, sunlight, and animal and plant biodiversity for the Magnolia Tianxiang Farm design project and produced relevant floor plans.

• Performed modelling analysis by combining GIS and Rhino.

• Simulated seasonal light, wind direction, and wind speed before selecting suitable habitats for plants and crops with different environmental requirements.

• Took full responsibility for the municipal square landscape design in Ding'an City, including site investigation and analysis, floor-plan design, and preparation of the design proposal.

SELECTED HONOURS & AWARDS

• Best Student Oral Presentation Award, TDWG 2026 Annual Conference, Oslo, Norway — 09/2026

• First Prize, Academic Scholarship, Shanghai Jiao Tong University — 10/2020

• Second Prize, Academic Scholarship, Shanghai Jiao Tong University — 10/2019

• Third Prize, National University Landscape Garden Design Competition, CHSLA — 10/2019

• National Academic Scholarship (Top 1%) — 11/2017

• Third Prize, National University Student Environmental Ecological Science and Technology Innovation Competition (Top 3%) — 09/2017

• Outstanding Graduate Student, Shanghai Jiao Tong University — 03/2021

• Outstanding Graduate Student, Hainan University — 2015–2017

ACADEMIC SKILLS

• *Ecological and biodiversity modelling*: biodiversity modelling, ecological response modelling, avian occurrence analysis, and ecosystem service assessment.

• *Geospatial analysis*: GIS, raster/vector processing, land-use/land-cover analysis, landscape metrics, LiDAR-derived vegetation metrics, and multi-source geospatial data integration.

• *Programming and statistical tools*: R, Python, MATLAB, Netica; statistical modelling, Bayesian modelling, reproducible workflows, and data visualisation.

• *Design and experimental tools:* Adobe Photoshop, Adobe Illustrator, SketchUp, Lumion, WUFI, Tobii eye-tracking equipment, and physiological-response data collection.

TEACHING & LEADERSHIP

• *Guest lecturer on urban biodiversity*, Master of Landscape Architecture programme, National University of Singapore — 2026.

• *Teaching Assistant for Academic Writing*, Integrity and Ethics and Landscape Garden Graduation Design — 09/2018–08/2020.

• *Leader*, Bulgarian Rural Research and Planning Practice Project, SJTU Global Challenge Program — 05/2019–08/2019.